Politecnico di Milano

Dipartimento di Elettronica, Informazione e Bioingegneria

prof. Fabrizio Ferrandi

Advanced Algorithms and Parallel Programming 1st part–February 14th 2025

Polimi ID _____ _____

Surname _____ _____ **Name** _____

- This is a closed-book examination. You cannot use computers, phones, or laptops during the exam.
- Paper will be provided, but you should bring and use writing instruments that yield marks dark enough to be read easily. Erasable pens can be used.
- Total available time: 1h:00m.

Exercise 1 (4 points) _____

Exercise 2 (4 points) _____

Exercise 3 (4 points) _____

Exercise 4 (4 points) _____

Exercise n. 1

Answer the following questions about parallel application modeling and ensure you provide a brief explanation. Remember, without a reason, the answer will be considered invalid. Understanding the questions is crucial for your success in this exam.

A. What can be said about the time complexity of a PRAM-based application when the PRAM model has a bounded number of processors? (1)

B. How is the communication via shared memory performed in a PRAM model? (latency vs parallel accesses)

C. Could you please discuss the differences between "weak scaling" and "strong scaling"? To which laws we have seen during the lectures are referring to? (1)

D. Please define and discuss the speed-up when Gustafson's law is considered. (1)

Exercise n. 2

Answer the following questions about advanced algorithms algorithms and briefly explain (without an explanation, the answer will be considered invalid)

A. Could you please describe how the powering algorithm works? Additionally, which class of algorithms does it belong to? (2)

B. Please discuss the role of the computed table on BDD ITE operation. How does the ITE asymptotic complexity change with this table?" (2)

Exercise n. 3

Answer the following questions about advanced algorithms and briefly explain (without an explanation, the answer will be considered invalid)

A. Could you explain which aspects of primality testing use randomization and how that process works? Additionally, would you classify it as a Las Vegas randomized algorithm? (2)

B. Describe how the split and the union work in a treap and discuss their complexity. (2)

Exercise n. 4

Answer the following questions about the algorithm analysis and briefly explain (without an explanation, the answer will be considered invalid)

A. Please discuss the lower bound of the comparison-based sort algorithms. (1) **This question can be skipped if you got 1 or 3 points in the first challenge.**

B. Please discuss the different variants of the amortized analysis. For what class of problems are they used? (2) **This question can be skipped if you got 2 or 3 points in the first challenge.**

C. Could you please list the characteristics of the counting sort algorithm that make it suitable for use in radix sort? (1)